Chapter 1

Study of the Existing Situation

1.1 Introduction

This chapter examines the existing situation of university activity management, analyzes current practices, and identifies the limitations of traditional methods. The analysis serves as a foundation for defining the problem and proposing a centralized digital solution to improve management processes and enhance participation.

1.2 University Activities Context

University activities constitute an essential component of higher education. They provide students with opportunities to engage in collaborative projects, develop professional skills, and explore interests beyond academic disciplines. Student clubs, in particular, serve as platforms for innovation and social interaction. These clubs may focus on technology, science, culture, or community service, enabling students to gain practical experience and strengthen their personal development.

Academic events such as conferences and workshops further contribute to knowledge dissemination and professional growth. Participation in these activities allows students and faculty members to present research, exchange ideas, and engage with emerging trends in their fields. Such experiences enhance critical thinking and encourage intellectual curiosity.

The organization of university activities involves multiple stakeholders, including students, club managers, professors, and administrative departments. Effective coordination among these actors is essential to ensure the success of events and initiatives. Clear communication channels and structured management processes are required to facilitate participation and resource allocation.

Despite their value, the increasing scale and diversity of university activities introduce organizational challenges. Traditional management methods often struggle to provide centralized information and efficient workflows. This situation highlights the need for structured solutions capable of improving coordination and accessibility within the university environment.

Actor	**Role**	**Description**
Students	Participants	Join clubs and events, enhance personal and professional skills
Clubs	Organizers	Create and manage activities, coordinate student participation
Administration	Supervisors	Approve events, manage resources, ensure proper governance
University	Enabler	Provides infrastructure, facilitates collaboration and engagement

Table 1.1: Roles of actors in university activities

1.3 Existing Management Methods

Currently, university activity management relies on decentralized and traditional approaches. Information about events and opportunities is typically shared through social media platforms, emails, posters, and informal communication channels. While these methods enable basic information dissemination, they lack centralization and consistency.

Registration processes for events and clubs are frequently handled through external tools or manual systems. Participants may complete online forms or physical registration documents that are not integrated into a unified platform. This fragmentation complicates data management and attendance tracking, limiting the ability to generate accurate statistics or insights.

Room reservation and resource allocation represent additional challenges. In many institutions, booking requests are processed through paper forms or email exchanges with administrative departments. These methods do not provide real-time visibility of room availability or automated confirmation mechanisms. As a result, scheduling conflicts and administrative delays often occur.

The absence of centralized communication and data management further reduces operational efficiency. Important announcements may be missed due to inconsistent dissemination practices. Additionally, fragmented data storage limits the ability to analyze participation trends or evaluate the effectiveness of activities.

Although traditional methods remain functional in basic scenarios, they are insufficient for modern university environments. The growing complexity of activity management requires structured and digital solutions capable of improving coordination and accessibility.

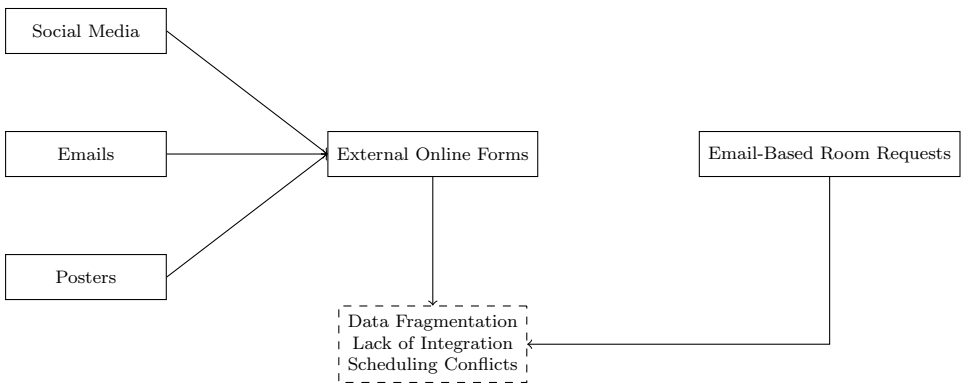

Figure 1.1: Fragmented Existing Activity Management Structure

1.4 Identified Problems and Limitations

The analysis of existing management practices reveals several key problems and limitations. The primary issue is the lack of centralization. Information related to clubs, events, and reservations is dispersed across multiple platforms, making it difficult for participants to access relevant details. This fragmentation reduces visibility and increases the likelihood of missed opportunities.

Limited accessibility of information represents another significant challenge. Students and staff may not receive timely updates about activities due to inconsistent communication methods. The absence of a unified information system limits participation and reduces engagement in university initiatives.

Inefficient registration and data management processes further hinder operational performance. Manual registration systems and external tools are prone to errors and do not integrate seamlessly with institutional workflows. This limitation complicates attendance tracking and prevents effective data analysis.

Room booking processes also suffer from inefficiencies. Traditional reservation methods do not provide real-time availability information or automated request handling. Scheduling conflicts and delays frequently occur, reducing the effectiveness of resource utilization.

Additionally, the lack of structured performance tracking limits decision-making capabilities. Universities require data on participation rates and activity outcomes to evaluate the success of initiatives. Without centralized data collection and analytics, it becomes challenging to identify areas for improvement or optimize resource allocation.

These limitations demonstrate that traditional management methods are insufficient for addressing the needs of modern university environments. A centralized and digital approach is necessary to enhance coordination, improve accessibility, and streamline administrative processes.

Problem	**Impact**
Decentralized information	Students may miss important events and opportunities
Manual registration	Data errors and inefficient attendance tracking
Room booking conflicts	Scheduling issues and administrative delays
Lack of analytics	Difficulty in evaluating activity performance

Table 1.2: Identified problems and their impacts

1.5 Proposed Solution

To address the challenges identified above, a centralized digital platform for managing university activities is proposed. The objective of this platform is to unify activity-related processes within a single system, providing structured workflows and improved communication among stakeholders.

The platform will enable students to explore clubs, register for events, and participate in activities through a unified interface. Club managers will be able to create and manage events, handle membership requests, and coordinate organizational tasks. Administrative staff will oversee announcements and approvals, ensuring proper governance of activities.

Room reservation processes will be digitized to provide real-time visibility of availability and simplify booking workflows. Users will be able to submit reservation requests, track their status, and receive confirmations through the platform. This approach eliminates manual procedures and reduces administrative overhead.

By centralizing data and automating workflows, the platform will enhance coordination and improve the overall management of university activities. It will provide transparency, streamline communication, and support data-driven decision-making. These improvements will contribute to a more organized and engaging university environment.

1.6 Conclusion

The findings of this chapter justify the need for further analysis and design of the proposed system. The next chapter will focus on system requirements, functional specifications, and modeling techniques to define the structure and behavior of the platform.

Chapter 2

Analysis and Design

2.1 Introduction

This chapter presents the theoretical and functional analysis of the proposed University Activity Management System. It begins with an overview of digital activity management systems, highlighting their objectives and benefits.

The functional and non-functional requirements are then specified based on the analysis of existing management practices within the university environment.

The Unified Modeling Language (UML) is introduced as a standard modeling tool used to visually represent system structure and behavior.

Finally, the system is modeled using UML diagrams, including use case diagrams, sequence diagrams, and class diagrams, to formalize interactions and define the architecture of the proposed solution.

2.2 Overview of University Activity Management Systems

University Activity Management Systems are digital platforms designed to support the organization and coordination of extracurricular and academic activities. These systems centralize information related to student clubs, events, workshops, conferences, and resource reservations, enabling structured communication among all stakeholders.

Traditional activity management methods often rely on manual or decentralized processes, which can lead to inefficiencies and information gaps. A centralized digital solution improves transparency, automation, and data accessibility.

2.2.1 Definition of Activity Management Systems

An Activity Management System (AMS) is an information system that facilitates the planning, execution, and monitoring of institutional activities.

In the university context, such a system enables:

- Management of student clubs and memberships.
- Creation and registration of events.
- Reservation of rooms and shared facilities.
- Distribution of announcements and notifications.
- Tracking and reporting of participation data.

By centralizing these processes, an AMS enhances operational efficiency and improves stakeholder collaboration.

2.2.2 Types of Activity Management Systems

Activity management approaches can be categorized as follows:

Manual Systems: Activities are managed using paper forms, emails, and informal communication. These systems lack integration and are prone to errors and delays.

Partially Digital Systems: Some processes are digitized using external tools such as online forms or social media platforms. However, data remains fragmented and unconnected.

Centralized Web-Based Systems: All activities and resources are managed within a unified platform. These systems provide role-based access, automated workflows, and real-time data management.

The proposed project adopts the centralized web-based approach to ensure efficiency and data consistency.

2.2.3 Objectives of a Digital Activity Management System

The primary objectives of implementing a digital system are:

- Centralizing activity-related information.
- Improving communication among students, faculty, and administrators.
- Automating registration and approval workflows.
- Enhancing transparency in event and room management.
- Providing statistical insights for performance evaluation.
- Increasing student participation and engagement.

These objectives align with modern digital transformation strategies in higher education.

2.2.4 Benefits of a Centralized Web-Based System

A centralized web-based solution offers significant advantages:

- **Efficiency:** Reduces administrative workload and processing delays.
- **Transparency:** Provides clear visibility of events and approvals.
- **Data Consistency:** Maintains a single source of truth for information.
- **Conflict Reduction:** Real-time availability prevents scheduling clashes.
- **Performance Monitoring:** Dashboards enable activity tracking and reporting.
- **Accessibility:** Users can access the system from any location and device.

These benefits contribute to improved coordination and resource management within the university.

2.3 Presentation of UML

The Unified Modeling Language (UML) is a standardized modeling language used in software engineering to visually represent system structure and behavior.

UML enables:

- Visualization of system architecture.
- Documentation of requirements and processes.
- Communication between technical and non-technical stakeholders.
- Blueprint creation for system implementation.

In this project, UML diagrams will be used to model user interactions, system workflows, and structural relationships.

2.4 Requirements Analysis and Specification

The objective of this phase is to analyze the context of university activity management and define system requirements based on identified limitations of existing methods.

This step ensures that the proposed solution addresses stakeholder needs and aligns with operational objectives.

2.4.1 Functional Requirements

The system shall:

- Allow user registration and secure authentication.
- Enable creation, modification, and deletion of clubs and events.
- Manage membership requests with approval workflows.
- Support room reservation requests and availability checks.
- Provide notifications for approvals and updates.
- Offer dashboards for activity monitoring and statistics.
- Implement role-based access control.

- Facilitate search and filtering of activities.
- Support data export in PDF or Excel formats.

These requirements define the core functionalities of the system.

2.4.2 Non-Functional Requirements

Technical Requirements

- The system shall be accessible at all times via a web interface.
- Authentication shall be required for user access.
- Data shall be protected against unauthorized access.
- Regular backups shall be performed to ensure data integrity.
- The system shall handle multiple concurrent users efficiently.

Usability Requirements

- The interface shall be intuitive and user-friendly.
- The system shall be compatible with desktops, tablets, and mobile devices.
- User actions shall provide immediate and clear feedback.
- The interface shall be professional and visually consistent.

These requirements ensure system reliability and user satisfaction.

2.5 Identification of Actors

The University Activity Management System involves multiple actors who interact with the platform based on their roles and responsibilities.

Clearly defining these actors allows for appropriate access control and workflow design.

2.5.1 Administrator

The Administrator oversees system configuration and global management.

Responsibilities:

- Manage user accounts.
- Approve or reject room reservations.
- Supervise announcements.
- Monitor system statistics.
- Oversee club and event management.

2.5.2 Student

The Student is the primary end-user of the platform.

Responsibilities:

- Register and log into the system.
- Browse clubs and events.
- Submit membership and event registrations.
- Receive notifications.
- View participation history.

2.5.3 Club Manager

The Club Manager is responsible for organizing and supervising club activities.

Responsibilities:

- Create and manage club information.
- Organize events.
- Process membership requests.
- Manage participant lists.
- Submit room reservation requests.
- View participation statistics.

2.5.4 Professor

The Professor organizes academic activities and requests resource reservations.

Responsibilities:

- Create academic events.
- Request room reservations.
- Monitor approval status.
- Access participation information.

2.6 System Modeling

The system modeling phase aims to formally represent the structure and dynamic behavior of the proposed University Activity Management System using the Unified Modeling Language (UML).

UML provides a standardized graphical representation of system components, their relationships, and interactions. In this project, the following diagrams are used:

- **Use Case Diagram:** Illustrates the interactions between system actors and the functionalities provided by the system.
- **Sequence Diagrams:** Describe the chronological flow of interactions between system components for specific scenarios.
- **Class Diagram:** Represents the static structure of the system, including entities, attributes, methods, and relationships.

These diagrams provide a clear and structured vision of the functional and technical architecture of the application.

2.6.1 Use Case Diagram

The general use case diagram presents the main interactions between system actors and the University Activity Management System. Each actor interacts with the system according to assigned privileges and responsibilities.

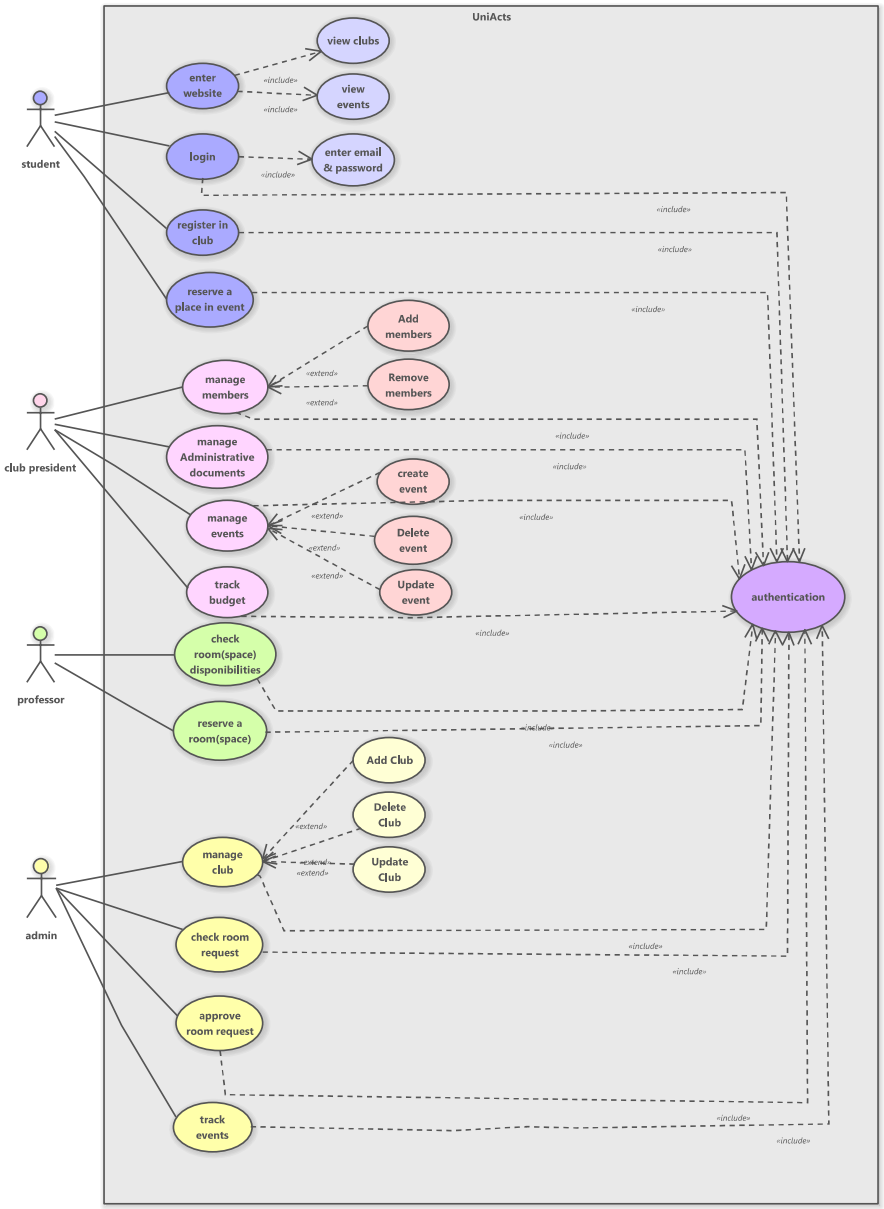

Figure 2.1: General Use Case Diagram

Textual Description of Use Cases

1. Textual Description – "Authenticate User"

Use Case Name	Authenticate User
Primary Actor	Student, Club President, Professor, Administrator
Description	Allows a registered user to access the system by entering valid credentials.
Preconditions	The user must have a registered account in the system.
Postconditions	The user is successfully logged into the system and redirected to the appropriate dashboard.
Main Scenario	1. User enters username and password. 2. System validates credentials. 3. System grants access and redirects to dashboard.
Alternative Scenario	1. Invalid credentials entered. 2. System displays error message.

Table 2.1: Textual Description "Authenticate User"

2. Textual Description – "Reserve Place"

Use Case Name	Reserve Place in Event
Primary Actor	Student
Description	Allows a student to reserve a place in a selected event.
Preconditions	The student must be authenticated. The event must have available seats.
Postconditions	The student is registered for the selected event.
Main Scenario	1. Student selects an event. 2. System checks seat availability. 3. Student confirms reservation. 4. System records the reservation.
Alternative Scenario	1. No seats available. 2. System displays notification message.

Table 2.2: Textual Description "Reserve Place"

3. Textual Description – "Reserve Room"

Use Case Name	Reserve Room
Primary Actor	Professor, Club President
Description	Allows authorized users to request a room reservation for an event or academic activity.
Preconditions	The actor must be authenticated. The requested room must be available.
Postconditions	A room reservation request is submitted and pending administrator approval.
Main Scenario	1. Actor selects room and date. 2. System checks availability. 3. Actor submits reservation request. 4. System records request as pending.
Alternative Scenario	1. Room unavailable. 2. System prompts actor to choose another date or room.

Table 2.3: Textual Description "Reserve Room"

4. **Textual Description – "Manage Club"**

Use Case Name	Manage Club
Primary Actor	Club President
Description	Allows the club president to manage club information, members, and activities.
Preconditions	The club president must be authenticated and assigned to a club.
Postconditions	Club information is updated successfully.
Main Scenario	1. Club President accesses club management page. 2. Updates club details or member list. 3. System saves changes.
Alternative Scenario	1. Invalid data entered. 2. System displays validation error.

Table 2.4: Textual Description "Manage Club"

5. Textual Description – "Manage Room Requests"

Use Case Name	Manage Room Requests
Primary Actor	Administrator
Description	Allows the administrator to review, approve, or reject room reservation requests.
Preconditions	There must be pending room reservation requests in the system.
Postconditions	The reservation request status is updated (approved or rejected).
Main Scenario	1. Administrator views pending requests. 2. Reviews reservation details. 3. Approves or rejects the request. 4. System updates reservation status.
Alternative Scenario	1. Missing or conflicting information detected. 2. Administrator requests modification or rejects request.

Table 2.5: Textual Description "Manage Room Requests"

2.6.2 Sequence Diagrams

Sequence diagrams represent the dynamic behavior of the system by showing message exchanges between actors and system components in chronological order.

Each object is represented by a lifeline, and messages illustrate method calls and data exchanges required to execute a specific scenario.

Sequence Diagram – Authentication

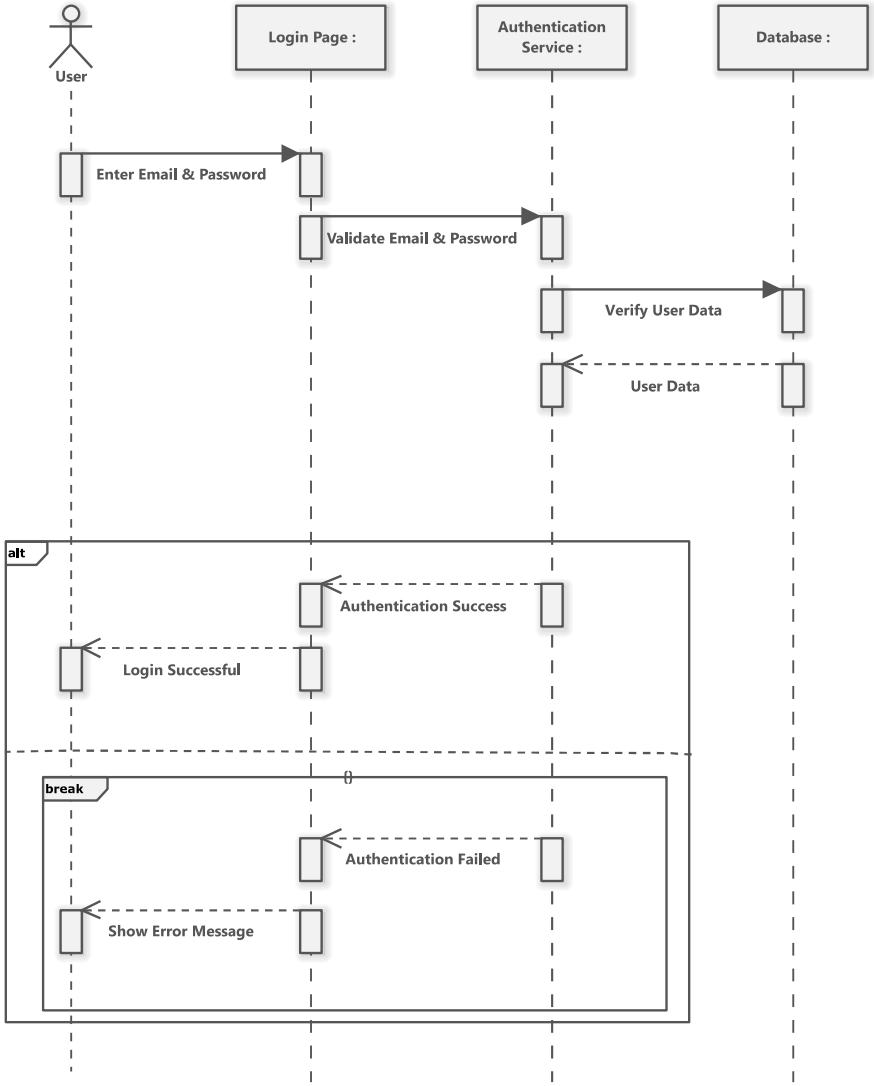

Figure 2.2: Sequence Diagram (Authentication)

Sequence Diagram – Reserve Place in Event

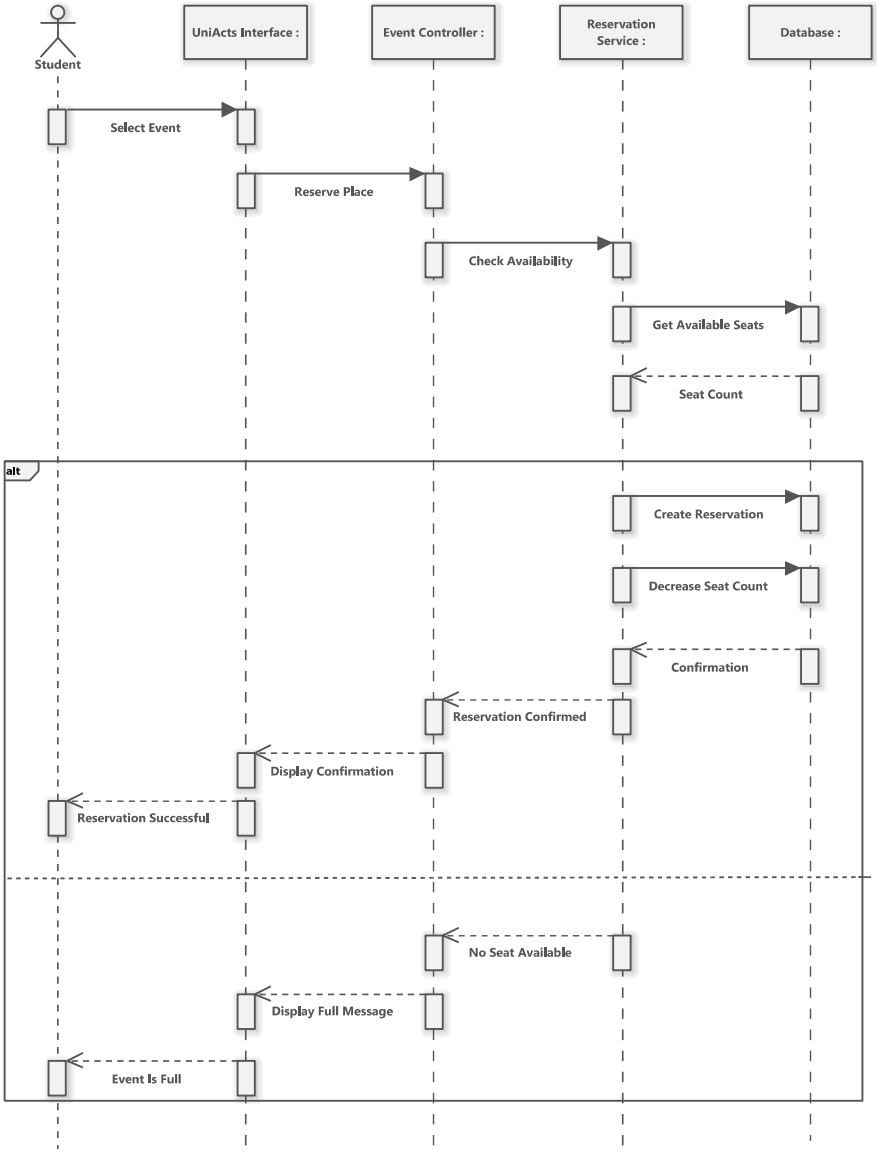

Figure 2.3: Sequence Diagram (Reserve Place in Event)

Sequence Diagram – Manage Club

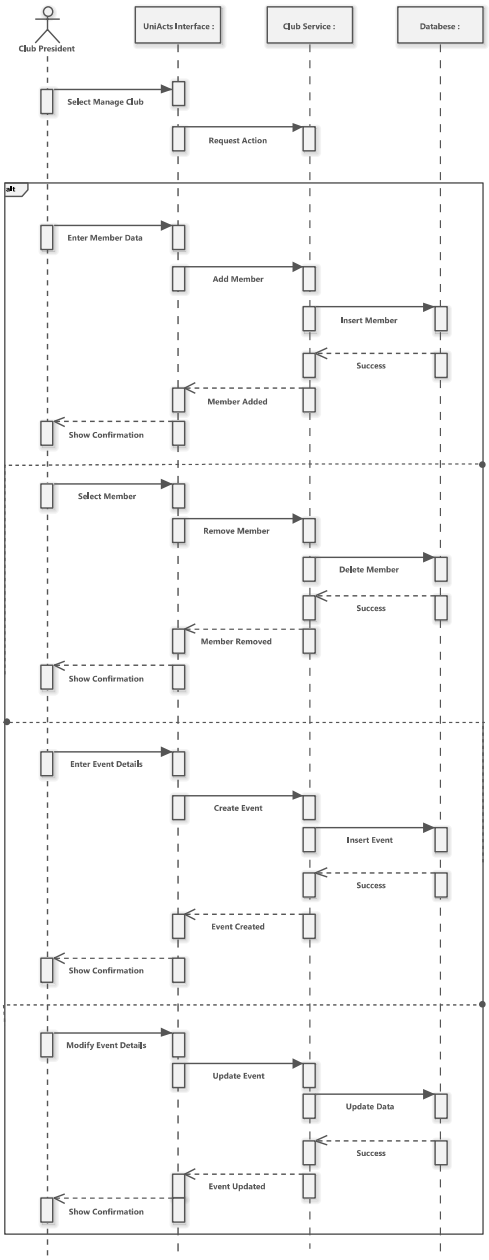

Figure 2.4: Sequence Diagram (Manage Club)

Sequence Diagram – Check and Reserve Room

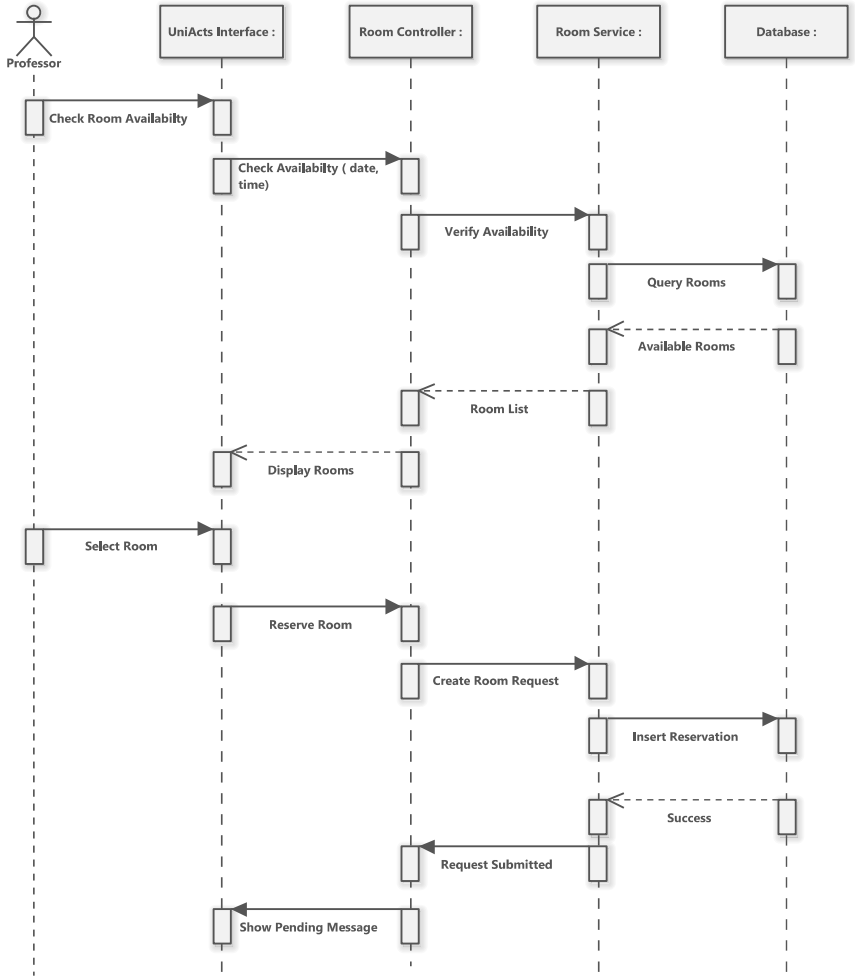

Figure 2.5: Sequence Diagram (Check and Reserve Room)

Sequence Diagram – Manage Rooms Reservations

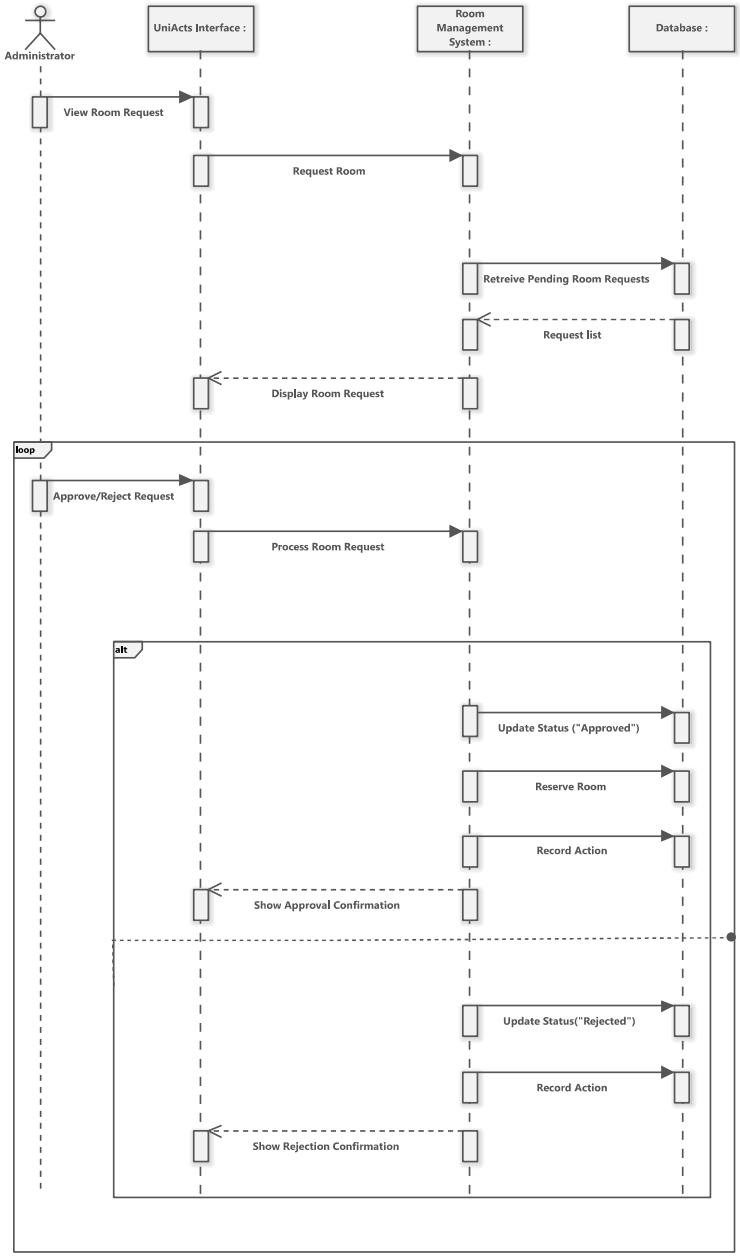

Figure 2.6: Sequence Diagram (Manage Rooms Reservations)

2.6.3 Class Diagram

The class diagram represents the static structure of the system and defines its main entities, attributes, operations, and relationships.

The system follows an object-oriented architecture centered around the abstract class *User*. This inheritance structure ensures role-based specialization while maintaining shared authentication behavior.

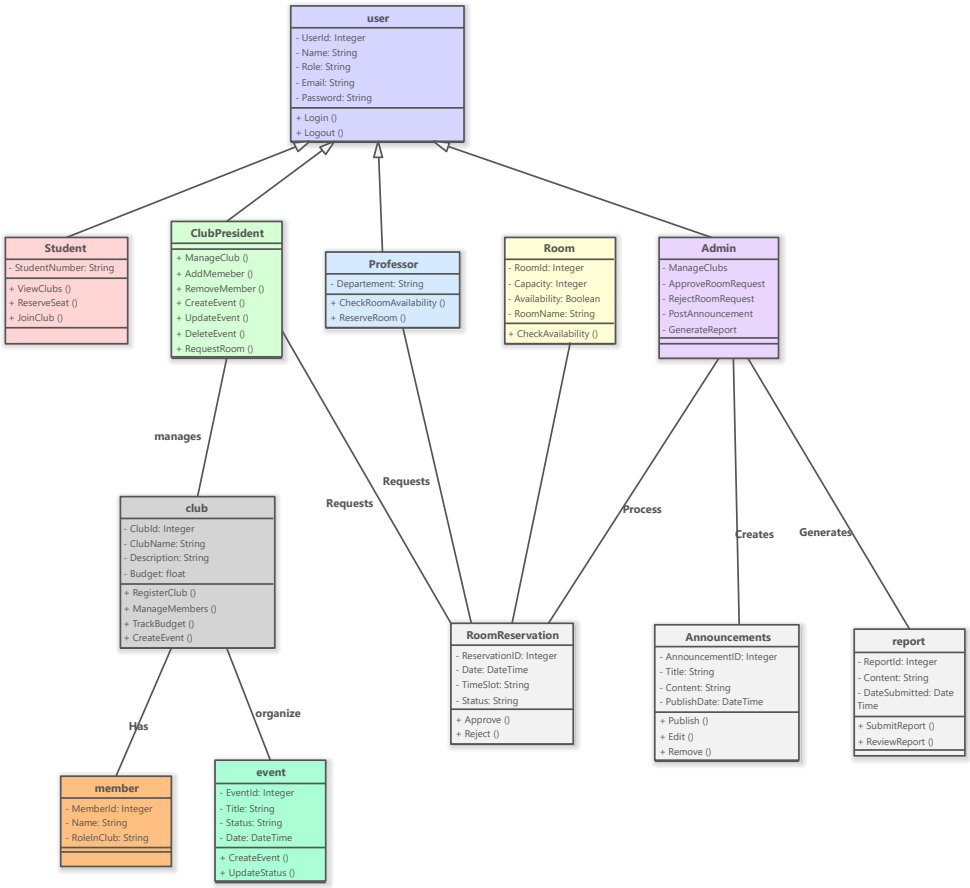

Figure 2.7: Classes Diagram

Management Rules

The system follows the following business rules:

- A User may have one specific role.
- A Club is managed by one Club President.
- A Club can have multiple Student members.
- A Club organizes multiple Events.
- A Student may register for multiple Events.
- An Event may have multiple registered Students.
- A Room cannot be reserved by more than one user for the same date and time slot.
- A Room Reservation is requested by either a Professor or a Club President.
- A Room Reservation is processed by an Administrator.
- An Administrator may generate multiple Reports.
- An Administrator may publish multiple Announcements.

These rules define system constraints and relationships.

2.6.4 Relational Model

The relational model of the University Activity Management System is defined as follows:

```
USER (user_id, name, email, password, role)

STUDENT (student_id, student_number, #user_id)

CLUB_PRESIDENT (president_id, #user_id)

PROFESSOR (professor_id, department, #user_id)

ADMINISTRATOR (admin_id, #user_id)

CLUB (club_id, name, description, creation_date, #president_id)

CLUB_MEMBER (#club_id, #student_id)

EVENT (event_id, title, description, date, time, capacity, available_seats, #club_id)

EVENT_RESERVATION (reservation_id, #student_id, #event_id)

ROOM (room_id, name, capacity, availability_status)

ROOM_RESERVATION (reservation_id, date, time_slot, status, #room_id, #user_id)

ANNOUNCEMENT (announcement_id, title, content, publish_date, #admin_id)

REPORT (report_id, generated_date, type, #admin_id)
```

2.7 Conclusion

This chapter provided a structured and formal representation of the proposed University Activity Management System.

Through requirement analysis, actor identification, and UML modeling, a clear understanding of system architecture and dynamic interactions was established. The use of use case, sequence, and class diagrams ensured both functional clarity and structural consistency.

This analysis and design phase establishes a solid foundation for the technical implementation phase, which will be presented in the following chapter.